

# The 2026-06-24 Sanriku-oki $M_w$ 6.9 earthquake (us6000t7zq): moment tensor and finite-fault analysis under interference from the Venezuela doublet

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## Abstract

We determine the point-source moment tensor (full and double-couple constrained, with a depth search) and a kinematic finite-fault model for the 2026-06-24 22:30:12 UTC  $M_w$  6.9 Sanriku-oki earthquake, for which USGS published moment tensors but no finite-fault model. The event occurred 25 minutes after the Venezuela M7.2+M7.5 doublet; we show quantitatively that the doublet’s global surface wavetrain sweeps 18 of 26 teleseismic surface-wave windows and 6 of 26 body-wave windows for this event, while W-phase/body-wave MT windows close before the interference arrives — a plausible operational explanation for the missing routine finite-fault product. After contamination-aware quality control, the moment tensor is robust ( $M_w$  6.90, interface thrust 184/24/65, centroid depth 46 km, in agreement with the USGS W-phase 177/31.5/60 at 45.5 km) and a marginal but coherent finite-fault model is recovered on the Japan Trench interface plane ( $M_w$  6.92, peak slip 1.3 m, 23–44 km depth).

## 1 Event and the interference problem

The  $M_w$  6.9 (Mww) thrust occurred at 40.2515°N, 142.1614°E (33 km ENE of Noda, Iwate), catalog depth 34 km, on the Japan Trench subduction interface. It followed the Venezuela M7.2 (22:04:33) + M7.5 (22:05:11) doublet by 25 minutes. Figure 1 shows, per station, the Sanriku body- and surface-wave analysis windows against the arrival of the Venezuela wavetrain (Love-wave onset to Rayleigh tail): 18/26 surface windows and 6/26 body windows are contaminated; only eight southern-azimuth stations retain clean surface waves. The W-phase and body-wave MT windows close by  $\sim$ 22:45, before the earliest interference ( $\sim$ 22:48), explaining why moment tensors were published while the long windows required by a finite-fault inversion are compromised.

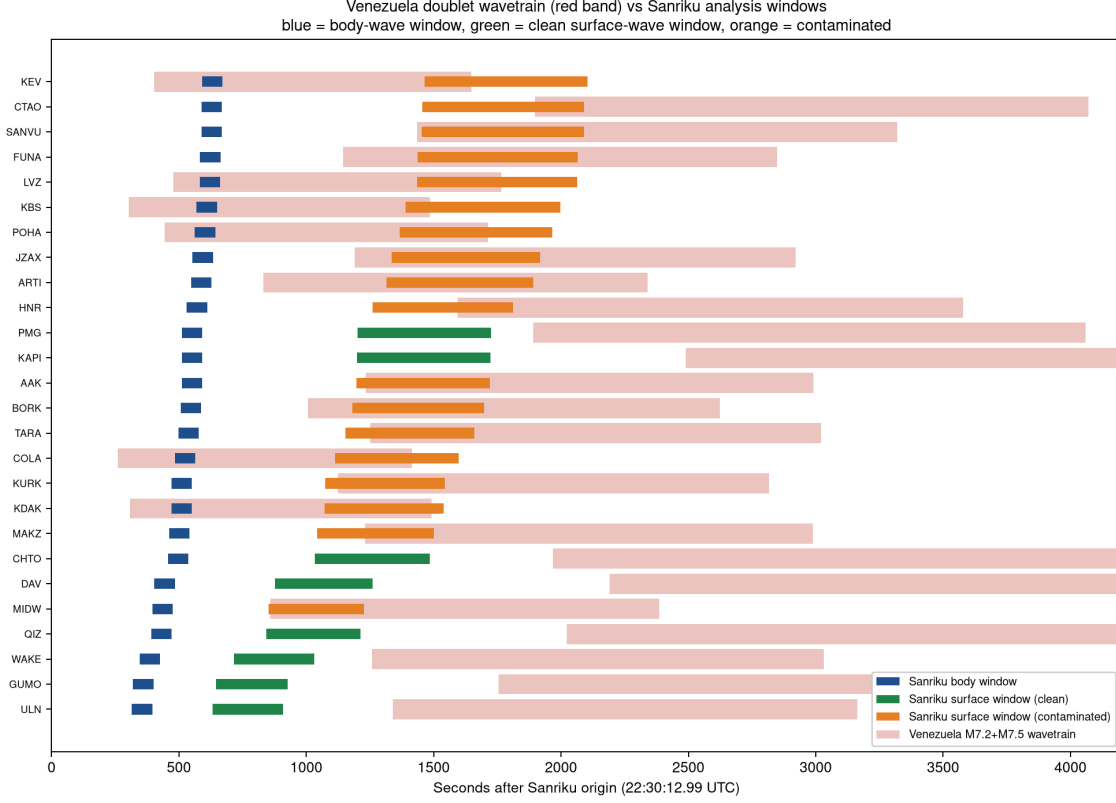


Figure 1: Venezuela doublet wavetrain (red) vs Sanriku analysis windows (blue = body; green/orange = clean/contaminated surface windows), seconds after the Sanriku origin.

## 2 Moment tensor (MTUQ)

Teleseismic GSN three-component data ( $25\text{--}75^\circ$ ), displacement, instrument response removed without a water level; body waves 10–33 s (20 clean stations), surface waves 50–100 s (8 clean stations); syngine ak135 Green’s functions; contamination encoded as CAP weights. Grid searches: full moment tensor (1.05M sources on the lune), double-couple (64k orientations  $\times$  5 magnitudes), and DC with depth as a search dimension (26–50 km).

| run        | $M_w$ | strike/dip/rake | depth         | note                     |
|------------|-------|-----------------|---------------|--------------------------|
| full MT    | 6.90  | 198/18/81       | 34 km (fixed) | thrust, interface family |
| DC         | 6.90  | 194/16/74       | 34 km (fixed) |                          |
| DC + depth | 6.90  | 184/24/65       | 46 km (best)  | clear misfit minimum     |
| USGS Mww   | 6.9   | 177/31.5/60     | 45.5 km       | reference                |

Table 1: Moment-tensor results. The depth search independently recovers the W-phase centroid depth; the mechanism converges toward the USGS solution as depth is freed.

Variance reduction on clean stations: surface waves 64.7%, body waves 8.5%. Body-wave shapes correlate well (cc 0.7–0.9) but synthetic amplitudes are systematically low, consistent with high-Q slab-path amplification unmodelled by 1-D ak135f; the mechanism is carried by the surface waves and body waveform shapes.

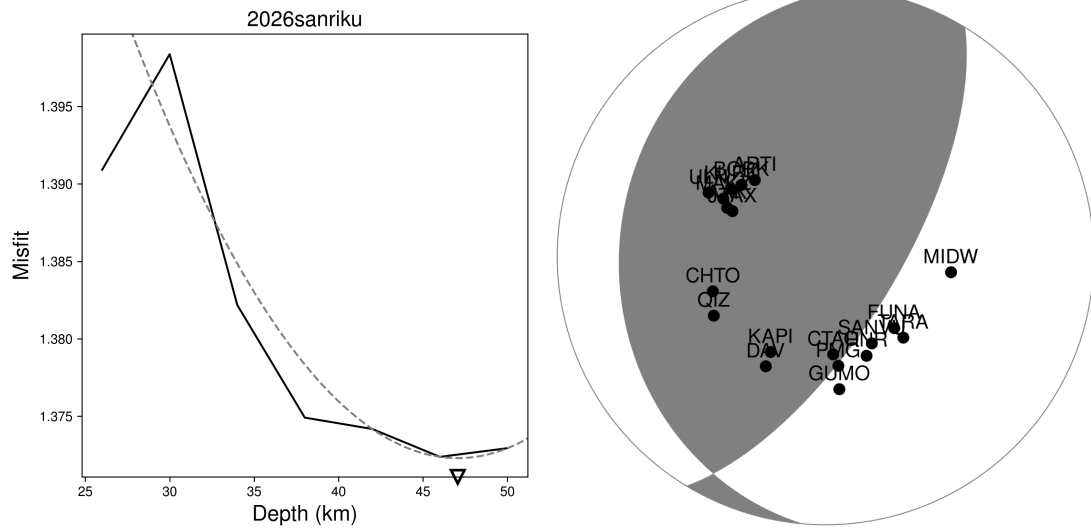


Figure 2: Left: DC misfit vs depth (minimum 46 km). Right: best DC (184/24/65) and stations.

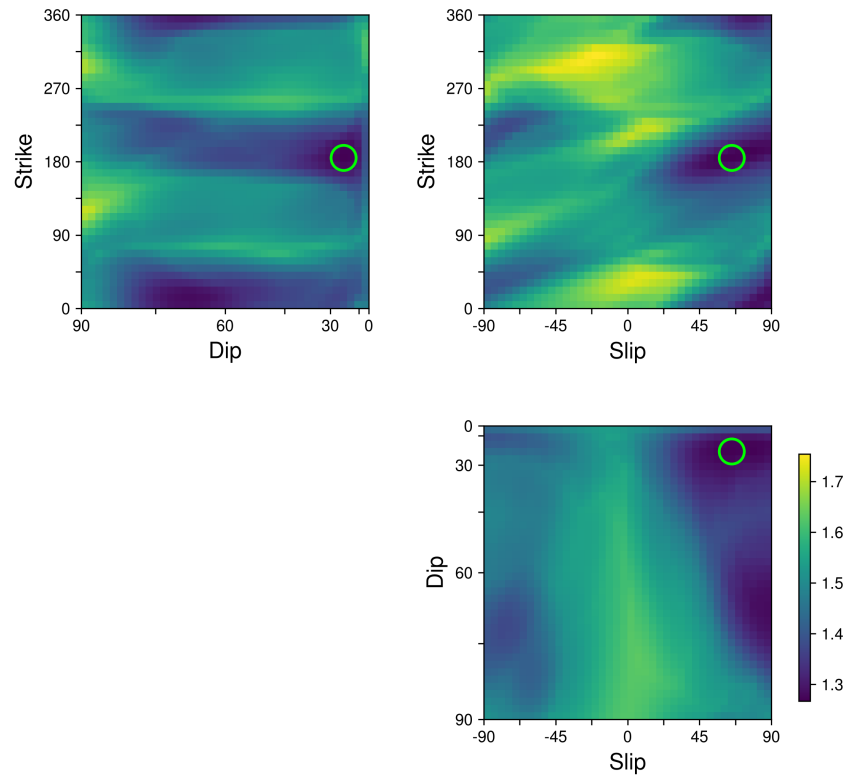
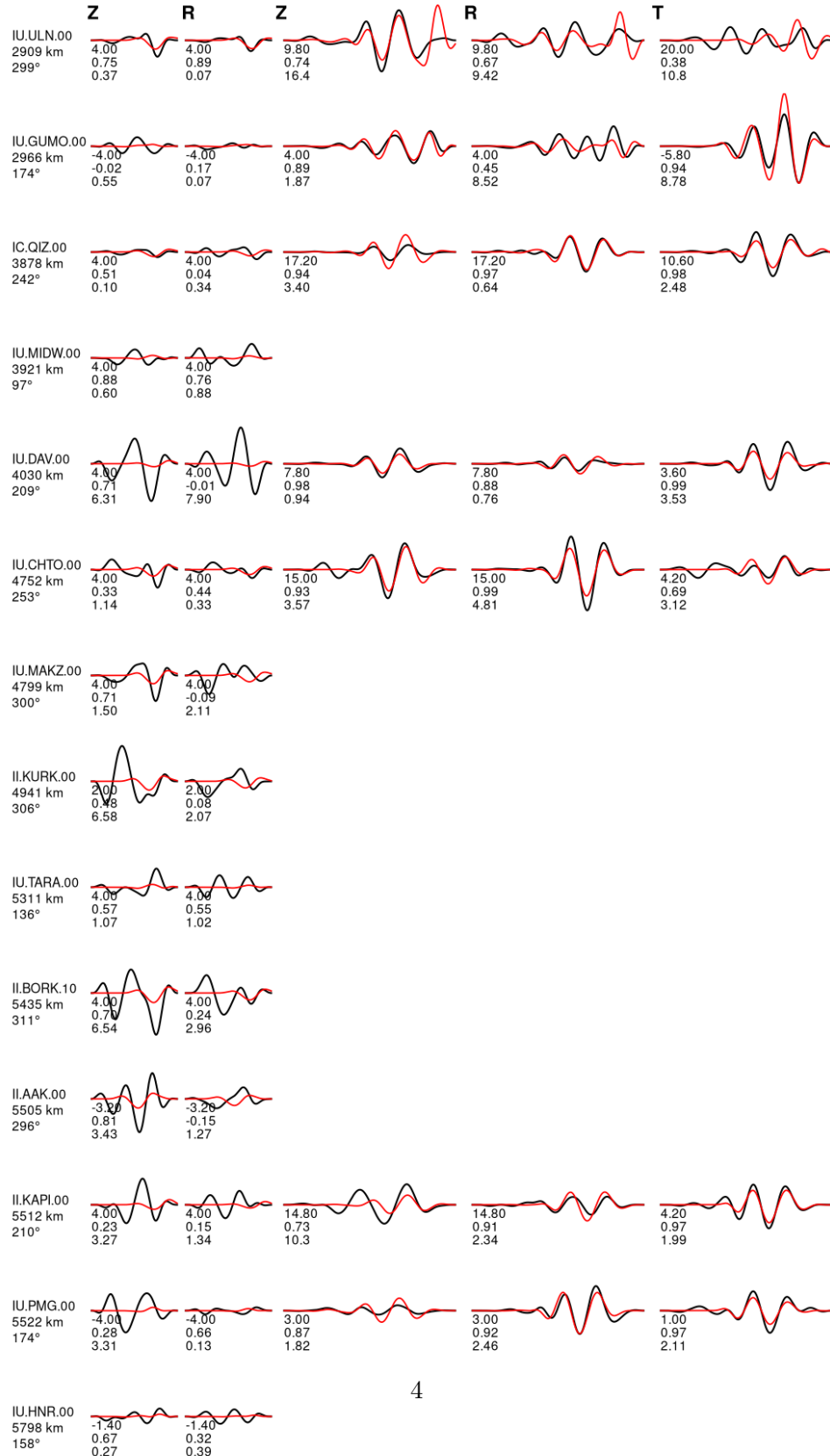


Figure 3: DC misfit surfaces (strike-dip-rake) at 46 km.



2026-06-24T22:30:12 40.25°N 142.16°E  $M_w$  6.90 Depth 46.0 km  
 model: ak135f\_2s solver: syngine misfit (L2): 1.267e+00  
 body waves: 10.0 - 33.3 s (40.0 s), surface waves: 50.0 - 100.0 s (300.0 s)  
 strike dip slip: 184 24 65, lune coords  $\gamma$   $\delta$ : 0 0, N-Np-Ns : 19-19-7



### 3 Finite-fault model (WISP)

Strict WISP CLI pipeline (auto model, both nodal planes; 54×44 km fault, both misfit 0.166,  $M_w$  6.91–6.92 — conjugate planes are teleseismically indistinguishable; the 31°-dipping interface plane NP2 (177/31) is the tectonically required choice). A key operational finding: WISP’s SNR-based station selection rejected all 74 processed P channels ( $\text{SNR} < 5$ ) — M6.9 teleseismic P sits at the microseism noise floor, independently corroborating that this magnitude is marginal for teleseismic finite-fault inversion. The refined model (NP2\_ref) force-includes 14 MTUQ-validated contamination-clean P stations, applies the contamination QC to SH and surface channels (8 SH + 3 surface survive), and re-inverts after cross-correlation alignment: final misfit 0.211.

| model               | plane  | $M_w$ | peak slip | depth extent | misfit             |
|---------------------|--------|-------|-----------|--------------|--------------------|
| auto NP1            | 32/63  | 6.91  | 1.37 m    | —            | 0.166              |
| auto NP2            | 177/31 | 6.92  | 1.32 m    | —            | 0.166              |
| NP2_ref (preferred) | 177/31 | 6.92  | 1.33 m    | 23–44 km     | 0.211 (clean data) |

Table 2: WISP finite-fault results.

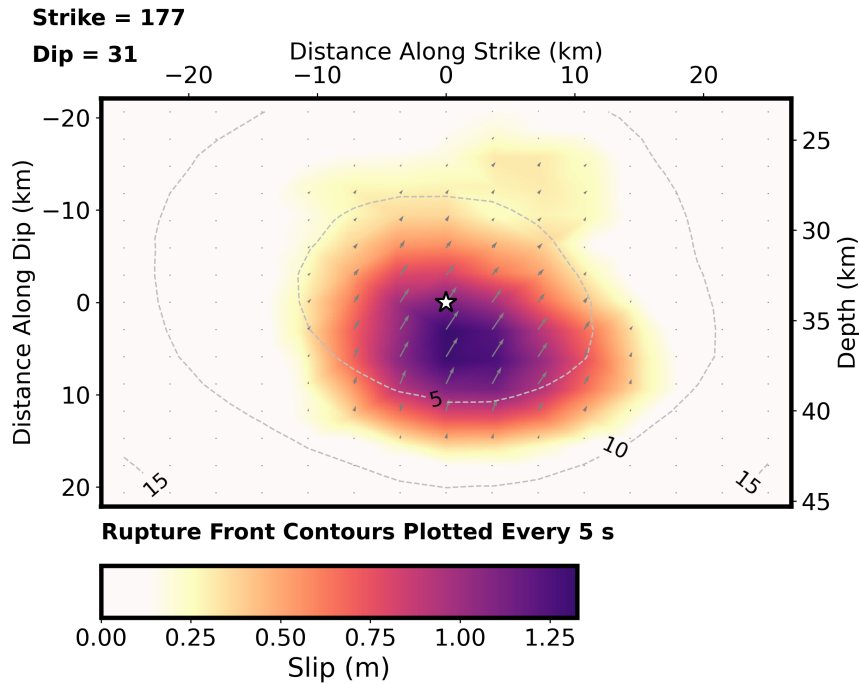


Figure 5: Preferred slip model (WISP native style, 5-s rupture-front isochrones).

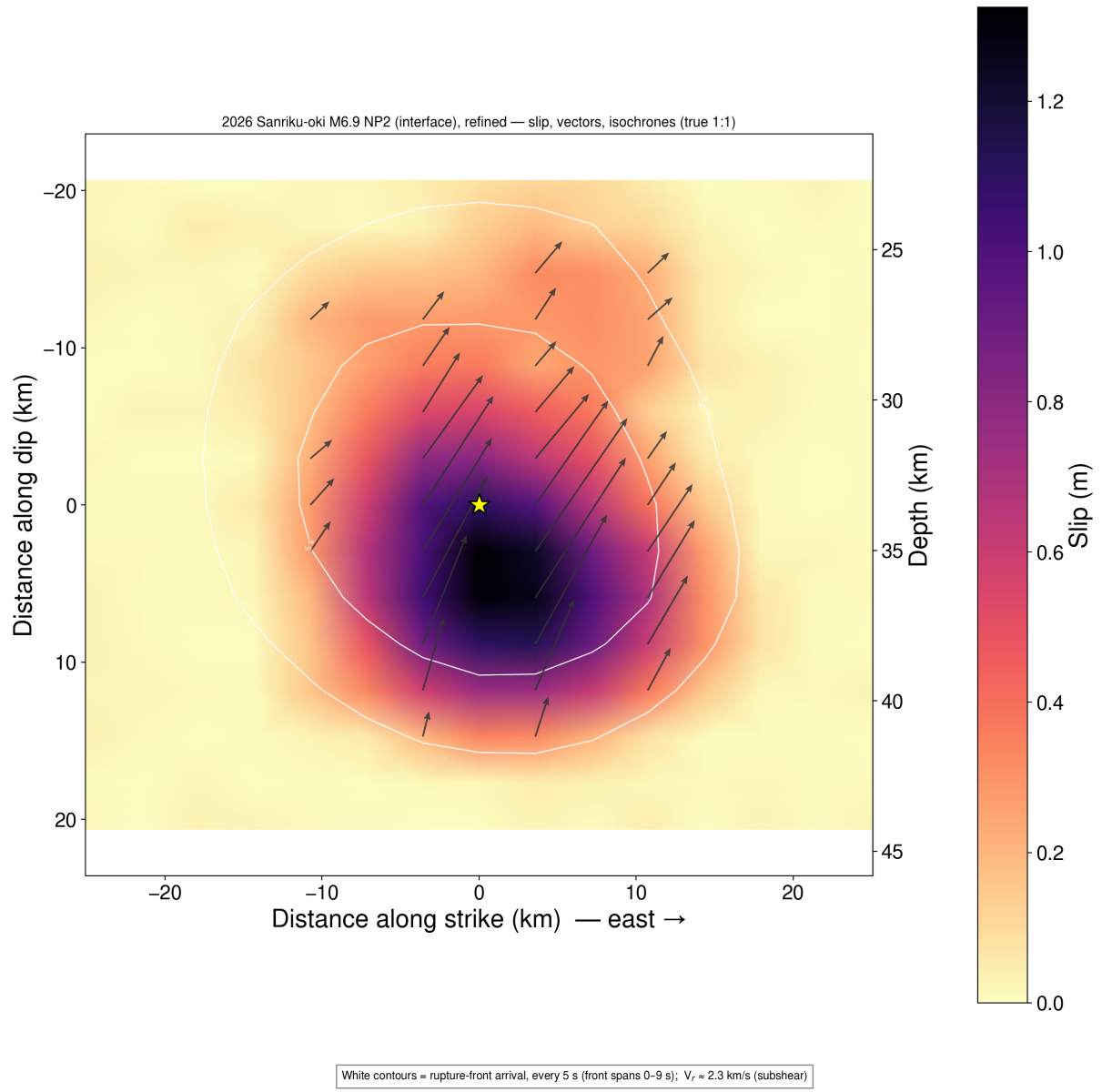


Figure 6: True-scale (1:1) slip, slip vectors, and isochrones on the interface plane.

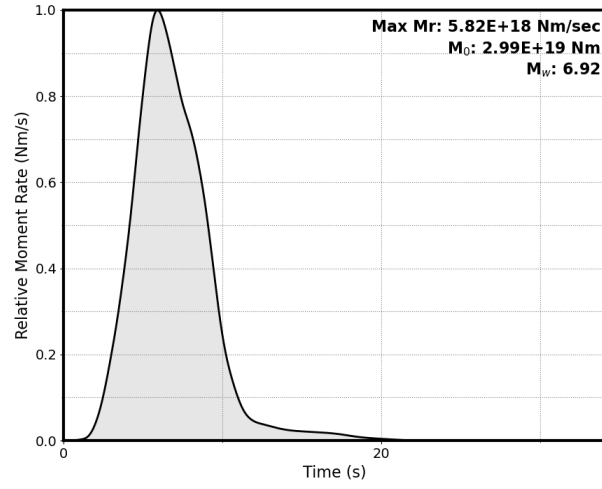


Figure 7: Moment-rate function ( $\sim 13$  s, consistent with the USGS source-time duration).

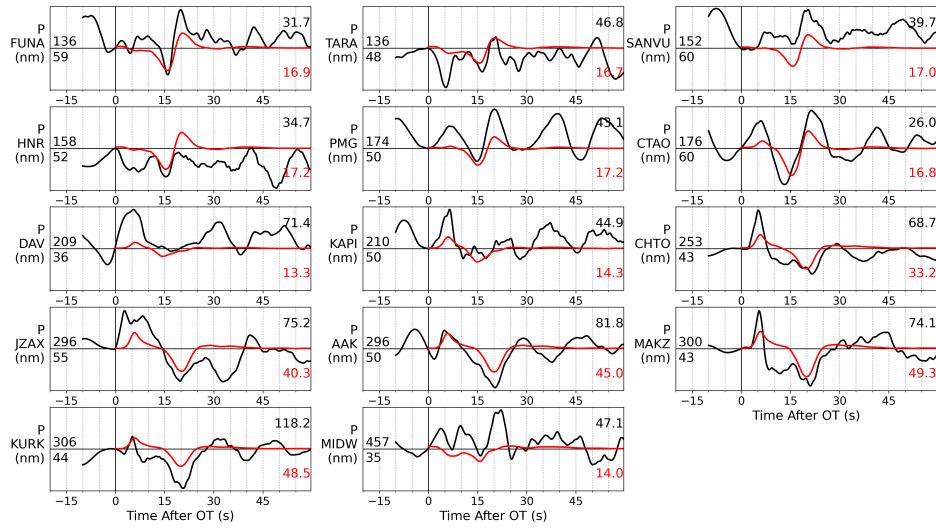


Figure 8: P-wave fits, refined model (faded = rejected by QC).

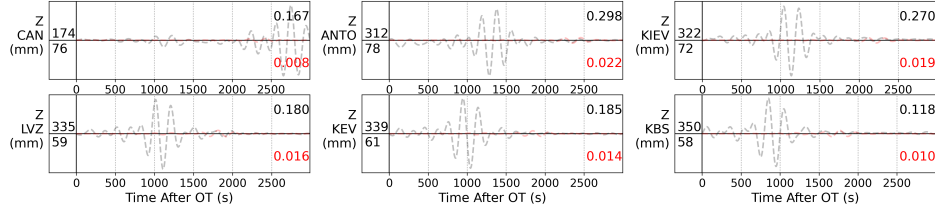


Figure 9: Rayleigh-wave fits (only contamination-clean channels are active).

## 4 Discussion and verdict

(1) Moment tensor: robust.  $M_w$  6.90 interface thrust within  $\sim 10\text{--}15^\circ$  of the USGS Mww; centroid depth 46 km independently recovered. The MT survives because its windows precede the interference. (2) Finite fault: obtainable but marginal, as predicted by the source-duration/usable-period criterion ( $T \approx 13$  s at M6.9): first-order content — a compact  $\sim 30\text{--}40$  km interface patch, peak  $\sim 1.3$  m, 23–44 km depth — is robust; internal detail, extent, and directivity are not resolved, and the plane choice is tectonic rather than data-driven. (3) Why USGS has no FFM for this event: the Venezuela doublet sweeps 69% of surface-wave and 23% of body-wave windows, and M6.9 P sits at the noise floor of operational SNR criteria; a routine FFM would face the same data starvation quantified here.